VINUNIVERSITY

**Test duration: 1 hour and 30 minutes**

**Name:** **Student ID:**

# Midterm Practice Test 4 – Solutions – Fall 2026

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**COMP3040/5030 Computer Vision** Unofficial practice test, prepared by Lê Tăng Phú Quý

## Instructions:

- Academic honesty is required in all work you submit to be graded. You should **NOT** copy or share your solutions with other students to avoid plagiarism issues.
- For short answer questions, please make your answers as **concise** as possible.
- This is a closed-book exam and NO Internet access is allowed. You are allowed 1 page of **hand-written** note (both sides).
- Drafting paper and pen/pencil are allowed. We will provide the drafting paper to you.
- Late submission without an approved extension will not be graded at all.
- You must **RETURN** this hard copy of the midterm exam at the end of the exam period.
- During the exam, you are **NOT** allowed to leave the room or/and use the restroom.
- You are required to **NOT** use your mobile phone in the exam room. Please put it on the table or in your bag.
- Good luck!

**About the exam**: There are **05** problems with **100 points** total, and some have multiple parts. Please check through the pages before you start.

| Problem | Content | Point (Undergraduates) | Type |
|---|---|---|---|
| 1 | Fundamentals of Computer Vision | 20 | short answer |
| 2 | Image Filtering | 25 | long answer |
| 3 | Feature Detection and Matching | 25 | long answer |
| 4 | Fitting and Image Transformations | 25 | long answer |
| 5 | System Design | 05 | long answer |
| **Total** | | **100** | |

## Problem 1. Fundamentals of Computer Vision

Give a short answer for each of the following question:

**1.1.** Give two examples of **geometric** and two of **semantic** information extractable from a street image. (5 pts)

*Solution.* Geometric: car contours/shapes, keypoint locations, camera pose, 3-D structure of buildings. Semantic: recognizing "car", "pedestrian", understanding the scene "intersection", reading a licence plate (OCR).

**1.2.** Distinguish **noise** from **outliers**. Which tools handle each? (5 pts)

*Solution.* Noise: small random deviations in every measurement (sensor) $\rightarrow$ smoothing, averaging, least squares. Outliers: grossly wrong values outside the distribution (dead pixels, wrong matches) $\rightarrow$ median, RANSAC, Hough, robust losses.

**1.3.** State the five properties of 2-D linear transformations (Lec07). Which one does translation violate? (5 pts)

*Solution.* Origin $\rightarrow$ origin; lines $\rightarrow$ lines; parallel lines stay parallel; ratios are preserved; closed under composition. Translation violates "origin $\rightarrow$ origin" (hence it is not linear).

**1.4.** Why does SIFT compare each DoG sample with **26** neighbours instead of just 8? (5 pts)

*Solution.* A keypoint must be an extremum in **both position and scale**: 8 neighbours in its level + 9 above + 9 below. Being extremal in scale gives the structure's characteristic scale $\sigma$ $\Rightarrow$ scale invariance.

## Problem 2. Image Filtering

**2.1.** (a) Prove discrete convolution is commutative: $(K*I)[n] = (I*K)[n]$. (b) $N$ images of the same static scene each carry independent zero-mean noise of variance $\sigma^2$. Prove the noise standard deviation of their average is $\sigma/\sqrt{N}$. (12 pts)

*Solution.* (a) $(K*I)[n] = \sum_m K[m]I[n-m]$; with $m' = n-m$: $= \sum_{m'} K[n-m']I[m'] = (I*K)[n]$. (b) $\mathrm{Var}\big(\frac{1}{N}\sum N_k\big) = \frac{1}{N^2}\sum \mathrm{Var}(N_k) = \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N}$ (cross-covariances vanish) $\Rightarrow$ SD $= \sigma/\sqrt{N}$.

**2.2.** Image $I = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$. (a) Define the integral image $S$ (with a leading zero row/column) and compute the entries needed to get the sum of rows 1–2, columns 1–2 with 4 lookups. (b) Why does an integral image make NSSD/NCC template matching fast? (13 pts)

*Solution.* (a) $S[y,x] = \sum_{y'<y,\, x'<x} I$. Sum $= S[3,3] - S[1,3] - S[3,1] + S[1,1] = 54 - 6 - 15 + 1 = 34\ (= 6+7+10+11\ \checkmark)$. (b) The denominators need $\sum P^2$ at **every** position: an integral image of $I^2$ gives each in $O(1)$ instead of $O(hw)$; the cross term $\langle T, P\rangle$ comes from a correlation/FFT.

## Problem 3. Feature Detection and Matching

**3.1.** (a) Prove the Harris response $R = \det M - k(\operatorname{tr} M)^2$ is unchanged when the image is rotated (assume a circular Gaussian window). (b) Explain why Harris is **not** scale-invariant, with an example. (13 pts)

*Solution.* (a) Rotating the image: $\nabla I' = R_\phi \nabla I \Rightarrow M' = \sum w\, R_\phi \nabla I \nabla I^\top R_\phi^\top = R_\phi M R_\phi^\top$ (a circular window is rotation-invariant). $\det$ and $\operatorname{tr}$ are invariant under orthogonal similarity $\Rightarrow$ same eigenvalues $\Rightarrow$ same $R$. (b) The window has a fixed size: zoom into a rounded corner and the old window sees only a gently curved segment $\approx$ an **edge** $\Rightarrow R < 0$; conversely shrinking can merge details into one corner. The size must adapt to the content (scale space).

**3.2.** (a) Why is the SIFT descriptor 128-dimensional? (b) Prove the descriptor (before clamping) is invariant to $I \to aI + b$, $a > 0$. (c) What is the purpose of clamping at 0.2 and renormalizing? (12 pts)

*Solution.* (a) A $16 \times 16$ region split into $4 \times 4$ cells, each an 8-bin orientation histogram: $4 \cdot 4 \cdot 8 = 128$. (b) $\nabla(aI + b) = a\nabla I$: $b$ vanishes, directions stay, magnitudes scale by $a \Rightarrow$ the histogram scales by $a \Rightarrow$ L2 normalization cancels $a$. (c) It limits the influence of a few very large gradients from **non-linear** illumination changes (saturation, specularities), emphasizing the distribution of orientations.

## Problem 4. Fitting and Image Transformations

**4.1.** (a) Derive the normal equations of least squares $\min_{\mathbf{w}} \|Y - X\mathbf{w}\|^2$. (b) For total least squares $\min \sum (ax_i + by_i - d)^2$, $a^2 + b^2 = 1$, prove the optimal line passes through the centroid and $(a, b)$ is the eigenvector of the smallest eigenvalue of the scatter matrix. (12 pts)

*Solution.* (a) $\nabla = 2X^\top X\mathbf{w} - 2X^\top Y = 0 \Rightarrow X^\top X\mathbf{w} = X^\top Y$ (convex $\Rightarrow$ global minimum). (b) $\partial/\partial d = 0 \Rightarrow d = a\bar{x} + b\bar{y}$; substituting gives $\mathbf{n}^\top S\mathbf{n}$ with $S$ the scatter matrix; Lagrange: $S\mathbf{n} = \lambda\mathbf{n}$, objective $= \lambda \Rightarrow$ pick $\lambda_{\min}$.

**4.2.** (a) Prove the composition of two affine maps is affine. (b) Find the inverse of $\mathbf{p}' = A\mathbf{p} + \mathbf{b}$. (c) With $A = \operatorname{diag}(2, 4)$, $\mathbf{b} = (4, -8)$, which point maps to $(10, 0)$? (d) Why does the shear $\begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$ keep every horizontal line in place? (13 pts)

*Solution.* (a) $A_2(A_1\mathbf{p} + \mathbf{b}_1) + \mathbf{b}_2 = (A_2A_1)\mathbf{p} + (A_2\mathbf{b}_1 + \mathbf{b}_2)$. (b) $\mathbf{p} = A^{-1}(\mathbf{p}' - \mathbf{b})$ (needs $\det A \neq 0$). (c) $(6/2,\ 8/4) = (3, 2)$. (d) $(x, y) \mapsto (x + 0.5y,\ y)$: $y$ is unchanged, so the line $y = c$ maps onto itself (points only slide along it by $0.5c$).

## Problem 5. System Design

**5.** An **augmented-reality** app: detect a specific book cover in phone video and overlay content exactly on the cover, even when the book is rotated, tilted, or partly covered by a hand. Design the system. (5 pts)

*Solution.* Offline: SIFT keypoints + descriptors of the canonical cover. Per frame: SIFT $\to$ NN matching + ratio test $\to$ RANSAC **homography** (planar cover $\Rightarrow$ 4 points, ~3 px threshold) $\to$ enough inliers $\Rightarrow$ book found; partial occlusion is fine because features are **local** and RANSAC only needs enough inliers $\to$ map the cover's 4 corners with $H$ for the overlay region, warp the

content with $H$ and blend. Speed & stability: track features between frames (optical flow) instead of re-detecting, smooth $H$ over time.